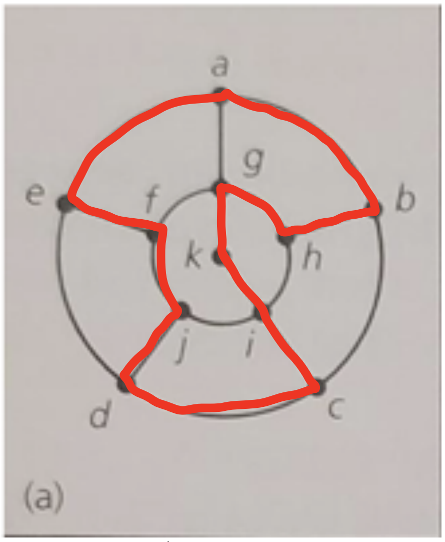
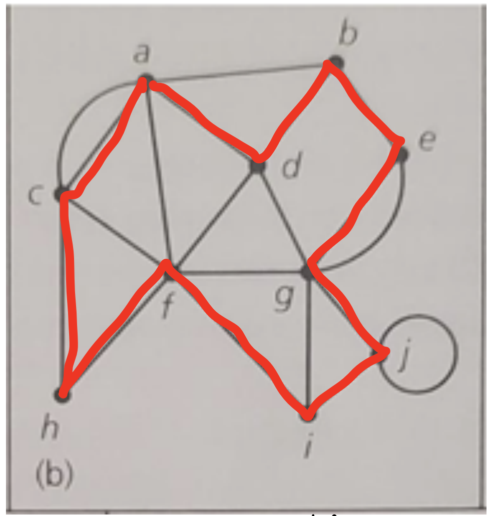
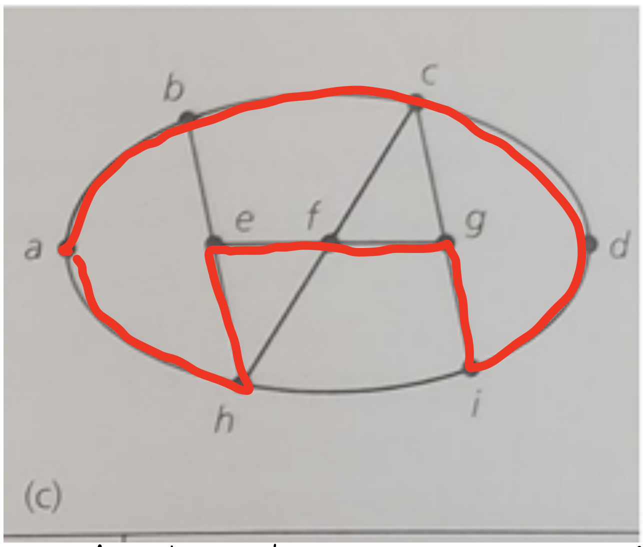
1.



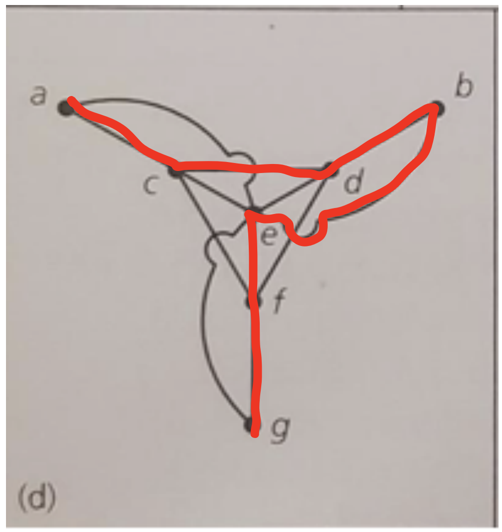
(a) has a Hamilton cycle: $a \to b \to h \to g \to k \to i \to c \to d \to j \to f \to e \to a$



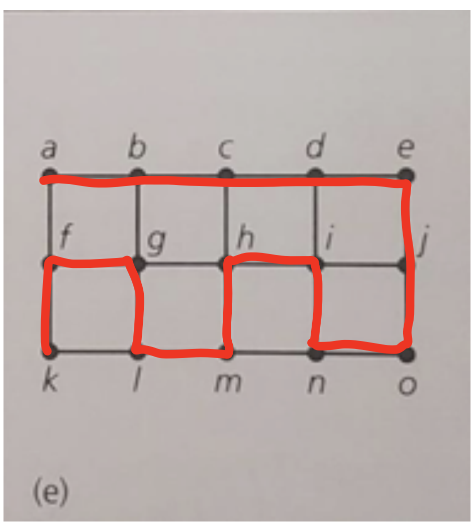
(b) has a Hamilton cycle: $a \to d \to b \to e \to g \to j \to i \to f \to h \to c \to a$



(c) has a Hamilton cycle: $a \to b \to c \to d \to i \to g \to f \to e \to h \to a$

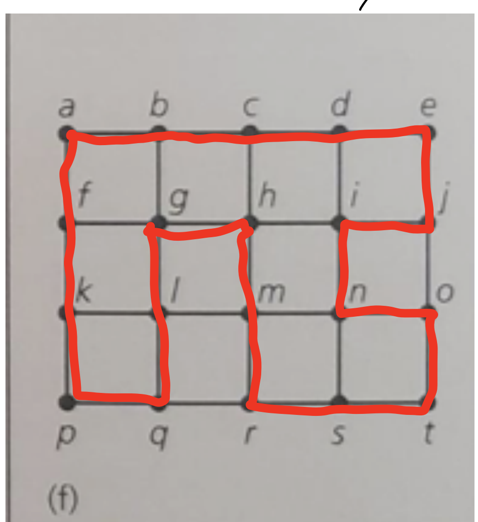


(d) has a Hamilton path: $a \to c \to d \to b \to e \to f \to g$ but no Hamilton cycle



(e) has a Hamilton path: $a \to b \to c \to d \to e \to j \to o \to n \to i \to h \to m \to l \to g \to f \to k$

but no Hamilton cycle



(f) has a Hamilton cycle: $a \to b \to c \to d \to e \to j \to i \to n \to o \to t \to s \to r \to m \to h \to g \to l \to q \to p \to k \to f \to a$

2. Since $(f(x))^2 = 2025 + \int_0^x ((f(t))^2 + (f'(t))^2)\,dt$

Let $h(x) = (f(x))^2 - 2025 - \int_0^x ((f(t))^2 + (f'(t))^2)\,dt = 0$

Of course $h'(x) = 0$

i.e. $2 f(x) \cdot f'(x) - (f(x))^2 - (f'(x))^2 = 0$

$\Rightarrow \quad (f(x) - f'(x))^2 = 0$

$\Rightarrow \quad f(x) = f'(x)$

Then solve the ODE problem

the general solution for $f'(x) = f(x)$ is $f(x) = Ce^x$ (C is a constant)

Consider the initial condition:

when $x=0$: $(f(0))^2 = 2025 + \int_0^0 ((f(t))^2 + (f'(t))^2)\,dt$

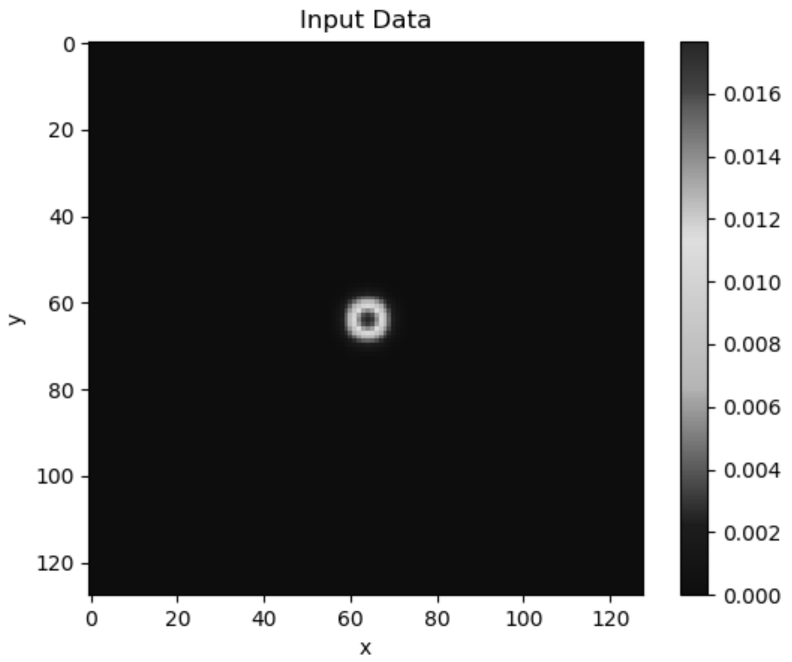
$\Rightarrow (f(0))^2 = 2025$

$f(0) = \pm 45$

plug it into $f(x) = Ce^x \Rightarrow C \cdot e^0 = \pm 45 \Rightarrow C = \pm 45$

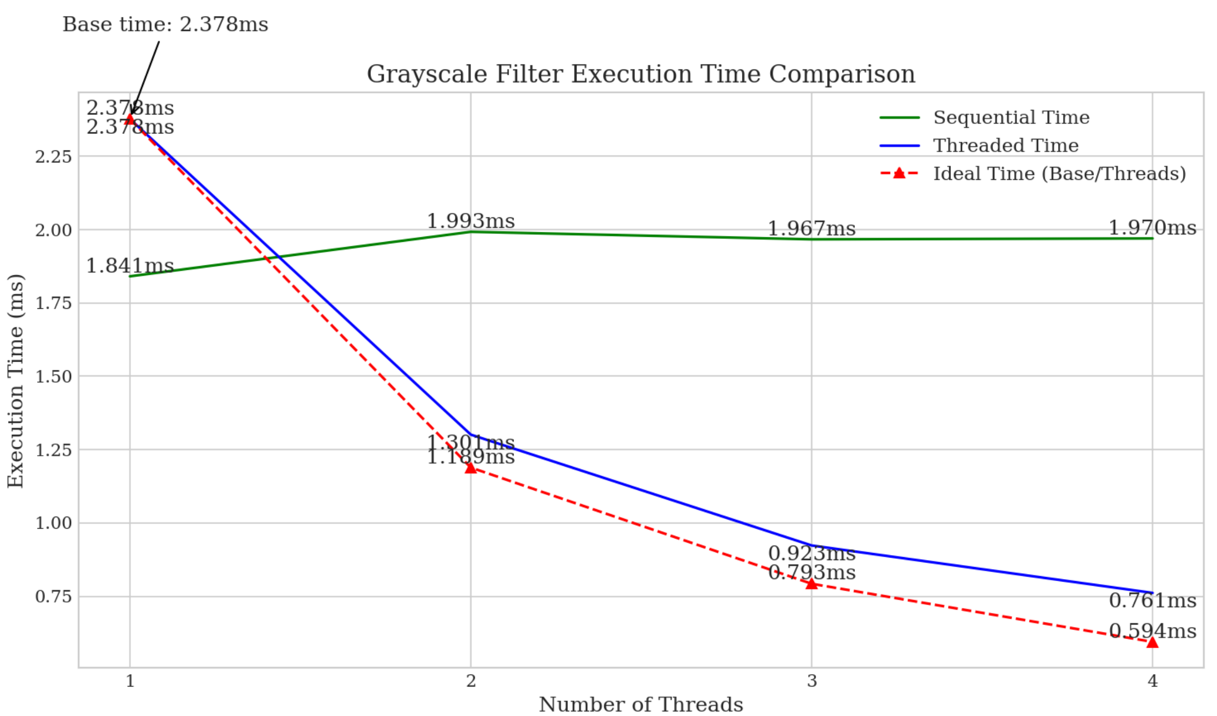
Thus, $f(x) = 45 e^x$ or $f(x) = -45 e^x$

3. (b)



Input Data
y
x
0
20
40
60
80
100
120
0.016
0.014
0.012
0.010
0.008
0.006
0.004
0.002
0.000

(c)



Base time: 2.378ms
Grayscale Filter Execution Time Comparison
Sequential Time
Threaded Time
Ideal Time (Base/Threads)
2.378ms
2.378ms
1.841ms
1.993ms
1.967ms
1.970ms
1.301ms
1.189ms
0.923ms
0.793ms
0.761ms
0.594ms
Execution Time (ms)
Number of Threads

4. Choose dimension = 4096